

Unit groups = $(\mathbb{Z}_m^{\times}, \times)$



To go from a group to a ring,

we need to add new operation.

Conversely, how does one get a group from a ring? One obvious answer

is to simply take the underlying abelian

group. That is, we just forget the

multiplication and focus on the addition

alone. But is there a group structure

with respect to the multiplication?

Def 1: let R be a ring. Define

$U(R)$ be the set of units in R .

eg. let $m > 0$. $R = \mathbb{Z}_m$.

Then $U(R) = \{ [a] : (a, m) = 1 \}$

Take $m = 8$. Then $\mathbb{Z}_8^\times = U(\mathbb{Z}_8)$

consists of $[1], [3], [5], [7]$.

Lemma 2: $U(R)$ is a group with

respect to multiplication of R .

eg. Consider $\mathbb{Z}_8^\times = \{ [1], [3], [5], [7] \}$.

To see $\mathbb{Z}_8^\times$ is closed under multiplication.

note that for $m > 0$, $(a, m) = 1$.

$(b, m) = 1$. then $(ab, m) = 1$.

Question: What is the size of $\mathbb{Z}_m^\times$?

Def 3: Let $m > 0$.

Define $\phi(m) = |\{1 \leq a \leq m : (a, m) = 1\}|$

This is called Euler's ϕ -function.

Lemma 4: Let p be a prime in $\mathbb{Z}$.

$$(1) \quad \phi(p) = p - 1$$

$$(2) \quad \phi(p^n) = p^n - p^{n-1} \quad \forall n \geq 1$$

proof : (1) is obvious, and

(2) $\Rightarrow$ (1) by setting $n=1$.

So it suffices to prove (2).

There are p^n numbers in total

from 1 to p^n . A number m is

not relatively prime to p^n iff m is

a multiple of p . There are p^{n-1}

multiples of p within 1 to p^n .

So $\phi(p^n) = p^n - p^{n-1}$.

lemma 5: let p, q be distinct primes

in $\mathbb{Z}$. Then $\phi(pq) = \phi(p) \phi(q)$

proof. Exercise.

lemma 6: let m, n be relatively prime.

Then $\phi(mn) = \phi(m) \phi(n)$

$$1 \leq s \leq mn$$

proof. let $A = \{1 \leq s < mn : (s, mn) = 1\}$

$$1 \leq u \leq m$$

$B = \{1 \leq u < m : (u, m) = 1\}$

$$1 \leq v \leq n$$

$C = \{1 \leq v < n : (v, n) = 1\}$

We construct a 1-1 correspondence

$$f: A \rightarrow B \times C.$$

Define $f: A \rightarrow B \times C$

$$s \mapsto (s \bmod m, s \bmod n)$$

We first show f is injective.

Suppose $\exists 1 \leq s' \leq mn$. $(s', mn) = 1$.

$$\text{s.t.} \quad s' \equiv s \pmod{m}, \quad s' \equiv s \pmod{n}$$

It follows $m \mid s' - s$ and $n \mid s' - s$

but $(m, n) = 1$. So $mn \mid s' - s$ (why?)

But $-mn < s' - s < mn$, So $s' - s = 0$.

That is, $s' = s$.

Now we show the surjectiveness.

Given $(u, v) \in B \times C$, we construct

$$1 \leq S < mn, \quad (S, mn) = 1 \quad \text{s.t.}$$

$$S \equiv u \pmod{m} \quad \text{and} \quad S \equiv v \pmod{n}.$$

Since $(m, n) = 1$, by Bezout's theorem,

$$\exists x_0, y_0 \in \mathbb{Z} \quad \text{s.t.} \quad mx_0 + ny_0 = 1.$$

Define $S = v m x_0 + u n y_0$

Note $S \equiv u n y_0 \equiv u \pmod{m}$

$$S \equiv v m x_0 \equiv v \pmod{n}$$

Since $mx_0 \equiv 1 \pmod{n}$, $ny_0 \equiv 1 \pmod{m}$

Cor T: let $m = p_1^{e_1} p_2^{e_2} \dots p_n^{e_n}$

for distinct prime p_i in $\mathbb{Z}$, and

$$e_i \geq 0 \quad \forall \quad 0 \leq i \leq n.$$

$$\text{Then } \phi(m) = \phi(p_1^{e_1}) \dots \phi(p_n^{e_n})$$

$$\text{eg. let } m = 8, \quad \phi(m) = \phi(2^3),$$

$$= 2^3 - 2^2 = 4, \quad \text{and } |\mathbb{Z}_8^\times| = 4.$$

$$\text{eg. Note in } \mathbb{Z}_8^\times, \quad [1]^2 = [1].$$

$$[3]^2 = [1], \quad [5]^2 = [1], \quad [7]^2 = [1].$$

$$\text{But in } \mathbb{Z}_9^\times = \{[1], [2], [4], [5], [7], [8]\}$$

$$\text{Here } \phi(9) = \phi(3^2) = 3^2 - 3 = 6.$$

we have $[4]^2 = [7] \neq [1]$.

So $\mathbb{Z}_8^\times$ is different from $\mathbb{Z}_9^\times$

not only in terms of size.

Def 8: let G be a group and $g \in G$.

The order of g is $|g| = \min \{ d > 0 : g^d = e \}$

Question: What is the order

of $[a]$ in $\mathbb{Z}_m^\times$?

eg. $|[8]| = 2$ in $\mathbb{Z}_9^\times$

$|[4]| = 3$ in $\mathbb{Z}_9^\times$

for $4^3 \equiv 7 \cdot 4 = 28 \equiv 1 \pmod{9}$.

Def 9: Let G be a group and

$g \in G$. Suppose $H \leq G$ is a subgroup.

The left coset of H with respect to g

$$is \quad gH = \{ gh : h \in H \}.$$

Lemma 10: Let G be a group and

$H \leq G$ a subgroup. Let $g \neq g'$ in G .

$$(1) \quad (a) \quad gH \cap g'H \neq \emptyset \quad \text{iff} \quad (b) \quad g^{-1}g' \in H$$

$$\text{iff} \quad (c) \quad gH = g'H$$

$$(2) \quad \forall g \in G, \quad |gH| = |H|$$

$$(3) \quad G = \bigsqcup_{g \in I} gH \quad \text{for some subset } I \subseteq G$$

proof. (i) $(a) \Rightarrow (b)$ because if

$x \in gH \cap g'H$, we write

$$x = gh = g'h' \text{ for some } h, h' \in H.$$

which implies $g^{-1}g' = h(h')^{-1} \in H$

$(b) \Rightarrow (c)$ Suppose $g^{-1}g' \in H$.

$$\text{Then } g' = g(g^{-1}g') \in gH$$

So $g'H \subseteq gH$. For $gH \cdot H \subseteq gH$.

Similarly, since $g^{-1}g' \in H$, its inverse

$$(g^{-1}g')^{-1} = (g')^{-1}g \in H. \text{ Then it follows}$$

$$g = g'(g')^{-1}g \in g'H. \text{ So } gH \subseteq g'H$$

(c) $\Rightarrow$ (a) obvious as $b \cdot h$.

gH and $g'H$ non zero because $e \in H$

and thus $g = ge \in gH$. $g' \in g'H$.

(2). We define $L_g : G \rightarrow G$

$$h \mapsto gh$$

This function has an inverse function

$$L_{g^{-1}} : G \rightarrow G \quad h \mapsto g^{-1}h$$

It is easy to check $L_g \circ L_{g^{-1}} = \text{id}_G$

and $L_{g^{-1}} \circ L_g = \text{id}_G$. So L_g is bijective.

Restrict L_g to H . we get $|H| = |\text{im } L_g| = |gH|$

Cor 11 : (Lagrange's theorem)

Let G be a finite group.

$H \leq G$ be a subgroup. Then $|H| \mid |G|$

Cor 12 : Let G be a finite group.

and $g \in G$. Then $|g| \mid |G|$.

Cor 13. (Euler's theorem)

Let $m > 0$. and $(a, m) = 1$.

Then $a^{\phi(m)} \equiv 1 \pmod{m}$.

Cor 14 : (Fermat's little theorem)

Let p be a prime and $a > 0$.

Then $a^{p-1} \equiv 1 \pmod{p}$.